Literature-Implied Status Note for arXiv:1401.4231

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Status. `literature_implied_answer`. This is an exact affirmative answer by a later theorem, identified during this run; it is not an original proof.

Source/open-problem metadata. Timur Oikhberg and Eugeniu Spinu, "Subprojective Banach spaces," arXiv:1401.4231, later published in J. Math. Anal. Appl. 424 (2015), 613–635. In the section "Tensor products of $C(K)$," immediately after the proof concerning $C(K)\widehat{\otimes}_\pi \ell_p$, the authors remark that if K is scattered and metrizable, they do not know whether

$$C(K)\widehat{\otimes}_\pi C(K)$$

is necessarily subprojective.

Supporting-answer metadata. R. M. Causey, "Subprojectivity of projective tensor products of Banach spaces of continuous functions," arXiv:2012.13440, posted 2020-12-24. Theorem 1 states that for every $n \in \mathbb{N}$ and compact Hausdorff spaces $K_1, \dots, K_n$, the projective tensor product

$$\widehat{\otimes}_{\pi,i=1}^{n} C(K_i)$$

is c_0-saturated if and only if it is subprojective if and only if each K_i is scattered.

Identification. Apply Causey's Theorem 1 with $n = 2$ and $K_1 = K_2 = K$. If K is scattered and metrizable, then it is a compact Hausdorff scattered space in the setting of the theorem. Therefore $C(K)\widehat{\otimes}_\pi C(K)$ is subprojective. Causey's theorem is stronger than the source remark because it does not require metrizability and treats all finite projective tensor powers.

Provenance limitation. The checked Causey source cites Oikhberg–Spinu for background on subprojectivity, but it does not appear to explicitly say that it is answering the specific 2014 remark. The packet is therefore classified as a literature-implied answer rather than a literature-already-answered packet. The implication itself is direct and answers the whole stated $C(K)\widehat{\otimes}_\pi C(K)$ question affirmatively.

Search evidence. The run indexes were searched for `1401.4231`, `2012.13440`, `Causey`, `Galego`, `Samuel`, `projective tensor product`, and `subprojectivity`. No prior packet for this exact target was found. Web search for the source remark surfaced arXiv:2012.13440, whose abstract states the relevant stronger projective tensor product criterion.

References

[1] T. Oikhberg and E. Spinu, "Subprojective Banach spaces," J. Math. Anal. Appl. 424 (2015), 613–635; arXiv:1401.4231.

[2] R. M. Causey, "Subprojectivity of projective tensor products of Banach spaces of continuous functions," arXiv:2012.13440, 2020.